17. Direct and Indirect Effects

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Direct and Indirect Effects* (Prof. Sanghack Lee, GSDS SNU), following Pearl (2001, UAI)

The central question is how to decompose a total causal effect into the part transmitted directly and the part transmitted through mediators, in nonlinear models where "holding the mediator fixed" and "subtracting" no longer give a well-defined indirect effect.

These notes consolidate the slides, define controlled, natural direct and natural indirect effects with nested counterfactuals, derive their identification formulas (including the mediation formula), discuss the cross-world problem, and introduce path-specific effects. Notation: treatment X (effect of $X = x$ relative to the reference $X = x^*$), response Y, mediators Z. Physical PDF page numbers are listed in Section C.

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1 Preliminaries: potential responses

Purpose. Fix counterfactual notation, including nested counterfactuals.

In an SCM, $Y(u)$ is the value of Y for unit $U = u$ (the potential response). $Y_x(u)$ is the value of Y for the same unit under $\text{do}(X = x)$, so $P(Y_x = y) = P(y \mid \text{do}(x))$. $Y_{xz}(u)$ is the value under the joint intervention $\text{do}(X = x, Z = z)$. $Z_{x^*}(u)$ is the mediator value that would naturally occur under the reference intervention $\text{do}(X = x^*)$. The nested counterfactual

$$Y_{x,Z_{x^*}}(u): \quad \text{set } X = x, \text{ but set } Z \text{ to the value it would have taken under } X = x^*.$$

2 Introduction

Purpose. Motivate direct and indirect effects and the prescriptive/descriptive distinction.

For $X \to Z \to Y$ with $X \to Y$:

- **Total effect:** the change in the distribution of Y when X is set to x by external intervention, $P(Y_x = y) = P(y \mid \text{do}(x))$.
- **Direct effect:** measured by holding all intermediate variables constant.
- **Indirect effect:** in a linear system, indirect = total − direct; in a nonlinear system it cannot be measured by holding some variables constant.

Informally, a direct effect quantifies an influence not mediated by other variables; more formally, the sensitivity of Y to changes in X while all other factors in the analysis are held fixed. Graphically, holding those factors fixed eliminates all causal paths from X to Y except the direct link $X \to Y$. Indirect effects cannot be defined this way: no choice of variables to hold constant makes the measured effect circumvent the direct pathway. Hence the definition of indirect effects remained incomplete.

2.1 Examples

Birth-control pill and thrombosis. The pill may increase thrombosis directly but reduce it indirectly by lowering pregnancy rates (pregnancy itself increases thrombosis). The direct effect is biologically stable and invariant to social factors (e.g. marital status), more transportable and more useful in scientific explanation and policy analysis than the total effect.

Race/sex discrimination in hiring. Courts care about the direct effect of race or gender on hiring, not the indirect effect via qualifications. The legal test: would the employer have made the same decision had the applicant's race or gender been different, all else being equal?

Smoking, hypertension and heart disease (Hernán and Robins 2020, Ch. 23). A randomised trial finds that quitting smoking ($A = 0$) reduces myocardial infarction Y. Does the benefit operate through reduced hypertension M (measured at 6 months)? Direct effect: $A \to Y$ not through M (oxidative stress, platelet effects); indirect effect: $A \to M \to Y$. Policy question: if a drug eliminated smoking-induced hypertension, how much of the MI benefit would remain?

Drug, aspirin and headache. A drug has a side effect, headache; patients with headaches tend to take aspirin, which may affect the disease directly or modify the impact of the drug. The total effect reflects existing aspirin usage patterns; the direct effect isolates the drug's impact regardless of aspirin intake. A direct effect such as

$$P(Y_{xz} = y) - P(Y_{x^*z} = y),$$

with z a specified aspirin level, informs whether aspirin should be encouraged or discouraged during treatment. In linear systems direct effects are given by path coefficients and do not depend on the level at which intermediates are held; in nonlinear systems that level generally modifies the effect and must represent the target policy. This leads to two conceptualisations.

2.2 Prescriptive versus descriptive

- **Prescriptive (controlled direct effect):** would an untreated patient improve if treated, while aspirin intake is held at a predetermined level $Z = z$? It does not rely on individual natural behaviour: $P(Y_{xz} = y) - P(Y_{x^*z} = y)$.
- **Descriptive (natural direct effect):** would the untreated patient improve if treated, while aspirin intake is held at the level the patient currently consumes under no treatment? It requires knowledge of individual natural behaviour.

Consider a patient who takes aspirin iff treated, and for whom treatment works only with aspirin. Then the natural direct effect is zero, whereas the controlled direct effect (at z = aspirin) is non-zero. For population averages: the prescriptive formulation is *pragmatic* (predict the difference in recovery when a prescribed aspirin dose is given to everyone), while the descriptive formulation is *attributional* (is an observed improvement due to the treatment itself, as opposed to preferential aspirin use among the treated?).

Why the average natural direct effect? Suppose the manufacturer considers eliminating the headache side effect. Would the drug retain its effectiveness in the population? The CDE does not answer this, since it refers to one aspirin level imposed uniformly. In the target population aspirin intake varies with factors other than drug-induced headache, which may also modulate effectiveness; hence the average natural direct effect is needed.

Descriptive indirect effects. There is no prescriptive definition of indirect effects, but the descriptive one transfers: to assess a patient's natural indirect effect, withhold treatment and ask whether the patient would recover if given as much aspirin as under treatment. (The prescriptive formulation cannot be transported, since the direct effect cannot be blocked by holding variables constant.) Their operational meaning concerns non-standard policies: if asking applicants about gender is made illegal (and no other indication of gender is available), any residual sex preference in hiring is attributable to the indirect effect of sex via qualifications. Its magnitude, predicted from data collected before the policy, is an average indirect effect.

3 Mathematisation

Purpose. Define controlled and natural direct effects and identify the latter.

Let X be the control variable, Y the response, Z the set of all intermediate variables between X and Y, and $Y_x(u)$ the value of Y for unit u under $\text{do}(X = x)$.

Definition 3.1 (Controlled direct effect)**.** $\text{CDE}_z(x, x^*; Y, u) = Y_{xz}(u) - Y_{x^*z}(u)$, and $\text{CDE}_z(x, x^*; Y) = \mathbb{E}[Y_{xz} - Y_{x^*z}]$, the expectation taken over u.

(The difference is the most common measure; other contrasts, e.g. ratios, can be used.)

Definition 3.2 (Natural direct effect)**.** $\text{NDE}(x, x^*; Y, u) = Y_{x,Z_{x^*}(u)}(u) - Y_{x^*}(u)$, and $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$.

3.1 The cross-world problem

$Y_{x,Z_{x^*}}$ refers to two intervention worlds at once: world 1 with $X = x$ (to evaluate Y) and world 2 with $X = x^*$ (to determine Z_{x^*}). No experiment, not even joint randomisation of X and Z, observes both worlds for the same individual. NDE and NIE are therefore empirically unverifiable in principle, and identification requires an untestable *cross-world independence* $Y_{xz} \perp\!\!\!\perp Z_{x^*}$ (Hernán and Robins 2020, §23.1; Robins and Richardson 2010).

3.2 Experimental identification

Formally, $\mathrm{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$ is not in general reducible to expressions of the form $P(Y_x = y)$ or $P(Y_{xz} = y)$. (Pearl's paper is from 2001; general nested-counterfactual identification results came later: Shpitser and Pearl 2008; Correa, Lee and Bareinboim 2021.)

Theorem 3.3 (Pearl 2001)**.** *If there is a set W of covariates, non-descendants of X or Z, with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$, the average NDE is experimentally identifiable:*

$$\mathrm{NDE}(x, x^*; Y) = \sum_{w,z} \big[\mathbb{E}(Y_{xz} \mid w) - \mathbb{E}(Y_{x^*z} \mid w)\big] P(Z_{x^*} = z \mid w) P(w).$$

Proof.

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid Z_{x^*} = z, w) P(Z_{x^*} = z \mid w) P(w) = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid w) P(Z_{x^*} = z \mid w) P(w),$$

$$\mathbb{E}[Y_{x^*}] = \mathbb{E}[Y_{x^*,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{x^*z} \mid w) P(Z_{x^*} = z \mid w) P(w),$$

using (in the first line) $Y_{xz} = Y_{x,Z_{x^*}}$ on the event $Z_{x^*} = z$ and the conditional independence, and (in the second) the composition axiom $Y_{x^*} = Y_{x^*,Z_{x^*}}$. Subtract. □

Graphical criterion. Example (a): $X \to Z \to Y$, $X \to Y$, and W a common cause of Z and Y. In (b), the subgraph $\mathcal{G}_{\underline{XZ}}$ obtained by deleting all arrows emanating from X and Z, W d-separates Y from Z: $(Y \perp_d Z \mid W)_{\mathcal{G}_{\underline{XZ}}}$, which ensures the condition.

3.3 Markovian models

The condition $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ holds with $W = \emptyset$ when Y_{xz} is independent of all variables in the model. In Markovian models:

- $P(Y_{xz} = y) = P(y \mid x, z)$, where $X \cup Z$ is the parent set of Y;
- $P(Z_{x^*} = z) = \sum_s P(z \mid x^*, s) P(s)$;
- $P(Y_{x,Z_{x^*}} = y) = \sum_s \sum_z P(y \mid x, z) P(z \mid x^*, s) P(s)$,

where S are the parents of Z other than X (or any set satisfying the back-door criterion). Hence the NDE is identifiable from non-experimental data:

$$\mathrm{NDE}(x, x^*; Y) = \sum_s \sum_z \big[\mathbb{E}(Y \mid x, z) - \mathbb{E}(Y \mid x^*, z)\big] P(z \mid x^*, s) P(s).$$

3.4 The mediation formula

In the simplest case ($X \to Z \to Y$, $X \to Y$, no confounding of Z–Y),

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]\, P(Z = z \mid X = x^*) :$$

evaluate Y under treatment x with the mediator distribution of treatment x^*. Smoking example (A treatment, M mediator): what would MI risk be if patients continued smoking ($a = 1$) but had the blood pressure of quitters ($a = 0$)?

$$\mathbb{E}[Y_{1,M_0}] = \sum_m \mathbb{E}[Y \mid A = 1, M = m]\, P(M = m \mid A = 0).$$

4 Natural indirect effects

Purpose. Define the NIE, relate it to the total effect, and identify it.

Definition 4.1 (Natural indirect effect)**.** $\mathrm{NIE}(x, x^*; Y, u) = Y_{x^*, Z_x(u)}(u) - Y_{x^*}(u)$, and $\mathrm{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*})$: keep the treatment at the reference level but let the mediator respond as under treatment.

Decomposition.

$$\begin{aligned}
\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}),\\
\mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}),\\
-\mathrm{NIE}(x^*, x; Y) &= -\mathbb{E}(Y_{x, Z_{x^*}}) + \mathbb{E}(Y_x).
\end{aligned}$$

Adding the last two lines gives

$$\mathrm{TE}(x, x^*; Y) = \mathrm{NDE}(x, x^*; Y) - \mathrm{NIE}(x^*, x; Y) :$$

the total effect is the direct effect of the transition $x^* \to x$ minus the indirect effect of the reverse transition. Strange as it looks, in the linear case it reduces to the standard additive relation $\mathrm{TE}(x, x^*; Y) = \mathrm{NDE}(x, x^*; Y) + \mathrm{NIE}(x, x^*; Y)$ (Appendix A).

Theorem 4.2 (Identification of the NIE)**.** *If there is a set W of non-descendants of X or Z with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ for all x and z, then*

$$\mathrm{NIE}(x, x^*; Y) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)\big[P(Z_x = z \mid w) - P(Z_{x^*} = z \mid w)\big] P(w).$$

In non-experimental studies the NIE is identifiable when $\mathbb{E}(Y_{x^* z} \mid w)$, $P(Z_x = z \mid w)$ and $P(Z_{x^*} = z \mid w)$ are. In a simple Markovian model,

$$\mathrm{NIE}(x, x^*; Y) = \sum_z \mathbb{E}(Y \mid x^*, z)\big[P(z \mid x) - P(z \mid x^*)\big].$$

(The proof mirrors the NDE case: $\mathbb{E}(Y_{x^*, Z_x}) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w) P(Z_x = z \mid w) P(w)$, and $\mathbb{E}(Y_{x^*})$ is the same with Z_{x^*}.)

5 Path-specific effects

Purpose. Generalise direct/indirect effects to effects along any chosen set of paths.

With several mediators ($X \to W \to Z \to Y$, $X \to Z$, $X \to Y$, ...) one may want the effect transmitted along a specific subset of paths. In the counterfactual graph, the inputs of each mechanism are switched between their values under x and under x^* according to whether the edge belongs to the selected paths.

Definition 5.1 (Path-specific effect; Pearl 2001)**.** Let $\mathcal{G}$ be the causal graph of $\mathcal{M}$ and g an edge-subgraph of $\mathcal{G}$ containing the paths selected for analysis. The g-specific effect of x on Y (relative to the reference x^*) is the total effect of x on Y in a modified model $\mathcal{M}_g$ built as follows. Partition each parent set as $Pa_i = \{Pa_i(g), Pa_i(\bar{g})\}$, where $Pa_i(g)$ are the parents linked to X_i in g and $Pa_i(\bar{g})$ the rest. Replace each $f_i(\mathbf{pa}_i, u)$ by

$$f_i(\mathbf{pa}_i, u; g) = f_i(\mathbf{pa}_i(g), \mathbf{pa}_i^*(\bar{g}), u),$$

where $\mathbf{pa}_i^*(\bar{g}) = Pa_i(\bar{g})_{x^*}$ are the values those variables would attain (in $\mathcal{M}$ and u) under $X = x^*$. Then $SE_g(x, x^*; Y, u)_{\mathcal{M}} = TE(x, x^*; Y, u)_{\mathcal{M}_g}$.

With $g = \{X \to Y\}$ this gives the NDE; with g the mediated paths, the corresponding indirect effect. Such effects lead to nested counterfactuals such as $Y_{W_x, Z_x, Z_{x^*}}$-type expressions, represented in counterfactual graphs.

Conclusions. Pearl (2001) defines path-specific effects by *path switching* rather than variable fixing, extending the interpretation and estimation of direct and indirect effects to nonlinear models. Under certain conditions these effects are identifiable even when Z is affected by X; the conditions always hold in Markovian models. The definition also gives an operational interpretation of indirect effects.

A The linear additive relation

Take $Z = \alpha X + U_Z$, $Y = \beta X + \gamma Z + U_Y$, so $Z_x = \alpha x + U_Z$ and $Z_{x^*} = \alpha x^* + U_Z$. Then

$$\begin{aligned}
\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}) = (\beta + \alpha\gamma)(x - x^*), \\
\mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}) = \beta(x - x^*), \\
\mathrm{NIE}(x, x^*; Y) &= \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*}) = \alpha\gamma(x - x^*),
\end{aligned}$$

so TE = NDE + NIE. (Also $\mathrm{NIE}(x^*, x; Y) = -\alpha\gamma(x - x^*)$, consistent with $\mathrm{TE} = \mathrm{NDE}(x, x^*) - \mathrm{NIE}(x^*, x)$.) This uses the linear, no-interaction form; nonlinear models do not generally preserve the same-direction sum.

B Counterfactual axioms

Composition. For singleton Y, W and any set X: $W_x(u) = w \Rightarrow Y_{xw}(u) = Y_x(u)$. Forcing a variable to the value it would have had anyway has no effect on other variables.
Consistency. $X(u) = x \Rightarrow Y(u) = Y_x(u)$.
Effectiveness. For all X, W: $X_{xw}(u) = x$.
Reversibility. For Y, W and any X: $(Y_{xw}(u) = y) \wedge (W_{xy}(u) = w) \Rightarrow Y_x(u) = y$.

C Source coverage

The source deck has 59 physical pages (37 numbered slides).

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